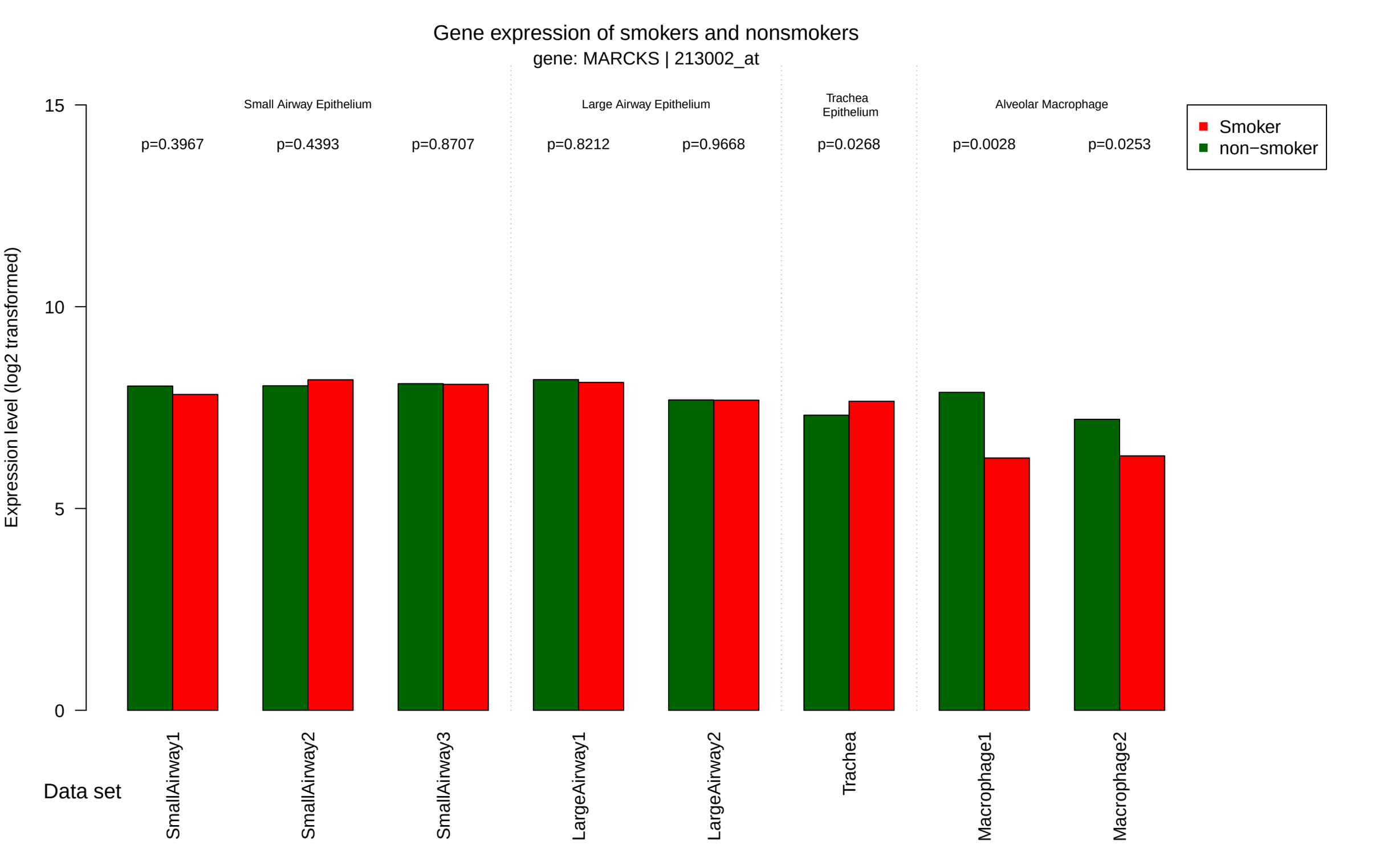
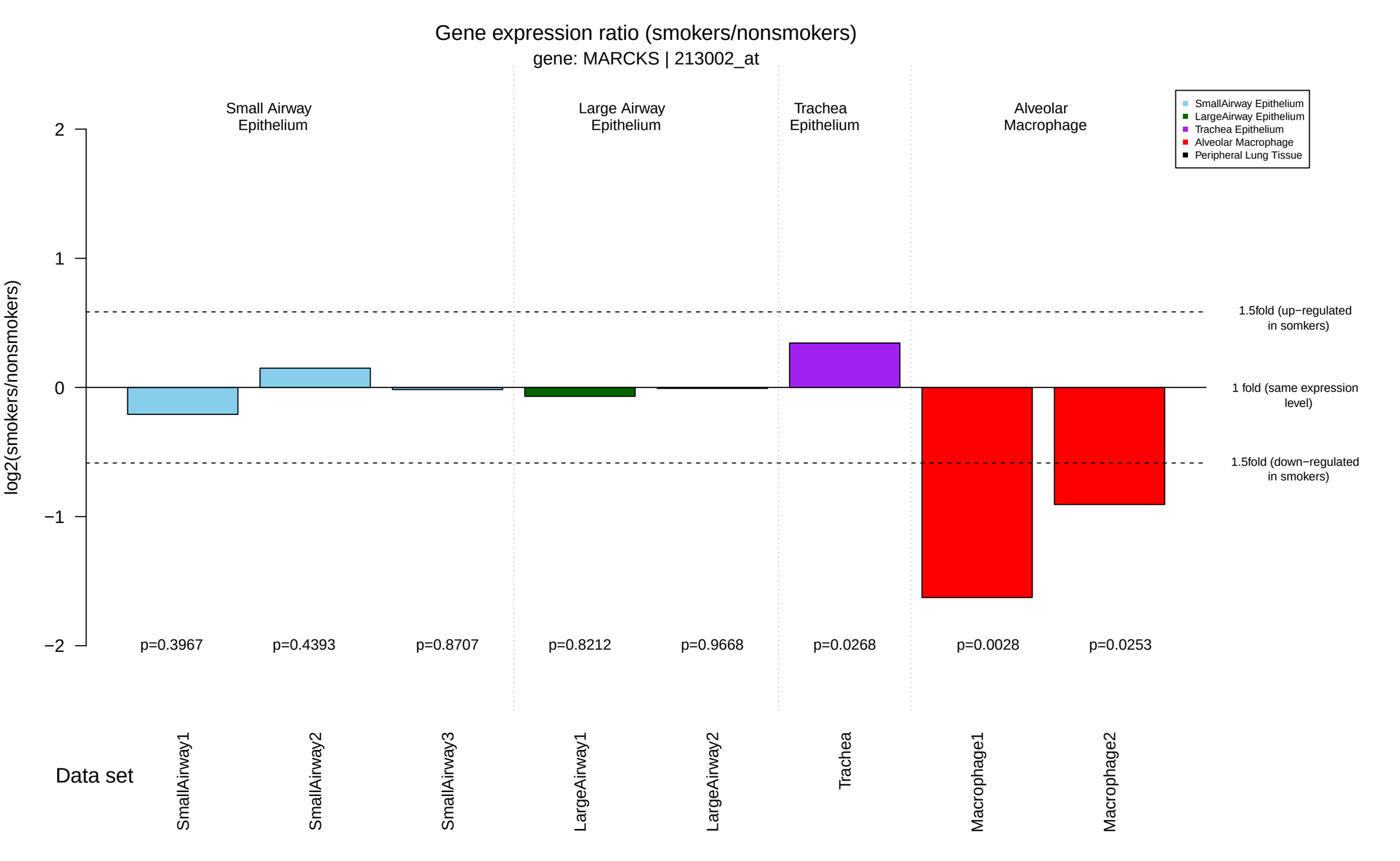
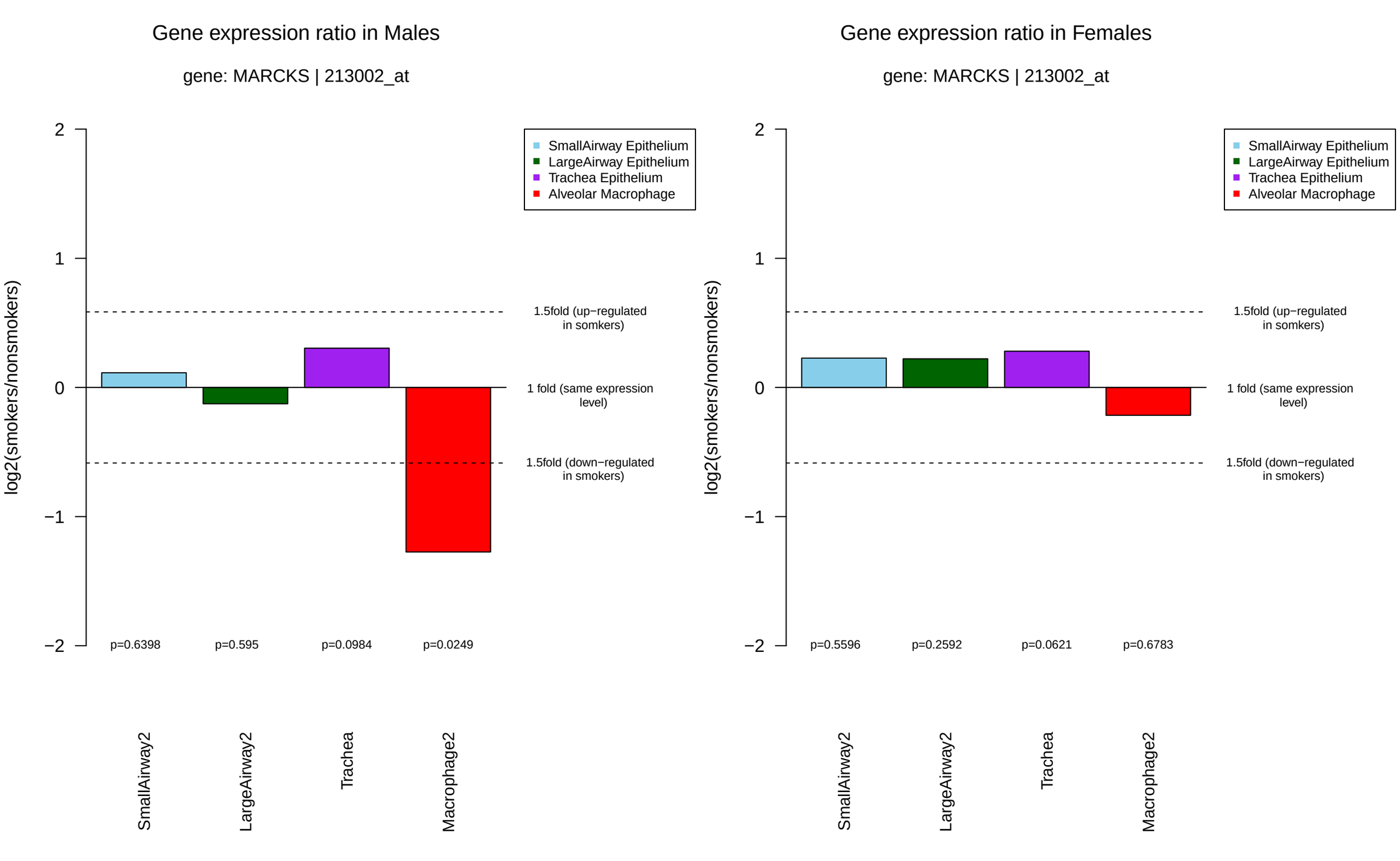
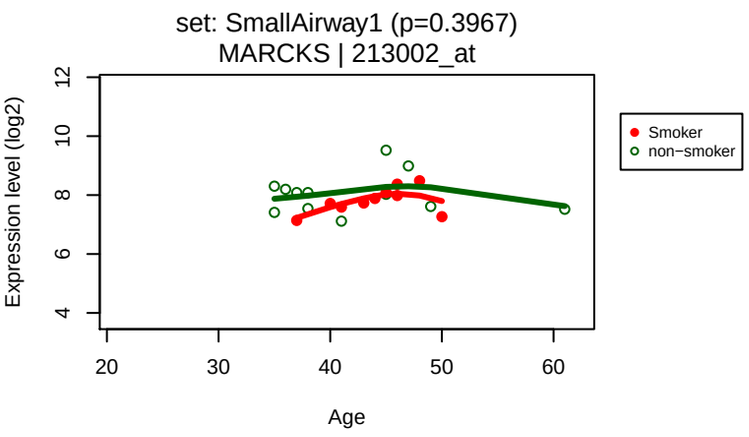
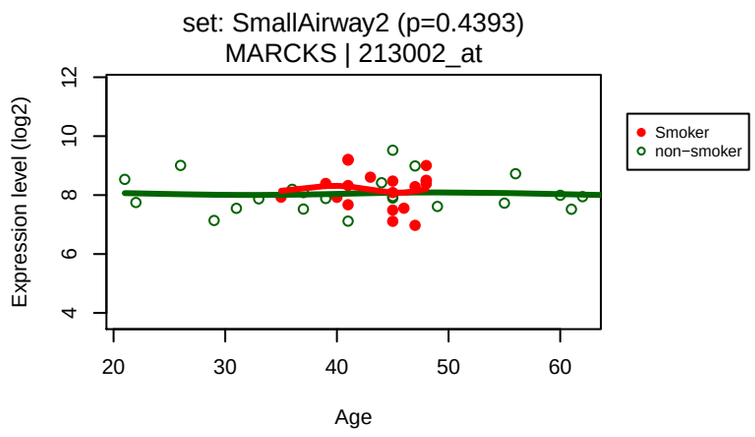
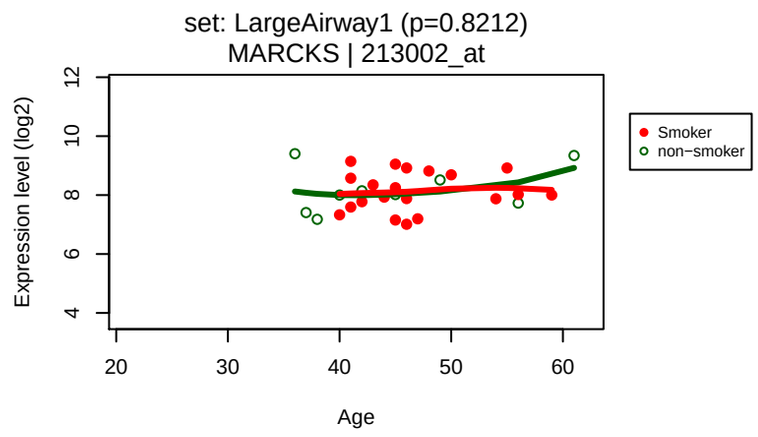

Gene expression of smokers and nonsmokers
gene: MARCKS | 213002_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3967
p=0.4393
p=0.8707
p=0.8212
p=0.9668
p=0.0268
p=0.0028
p=0.0253
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

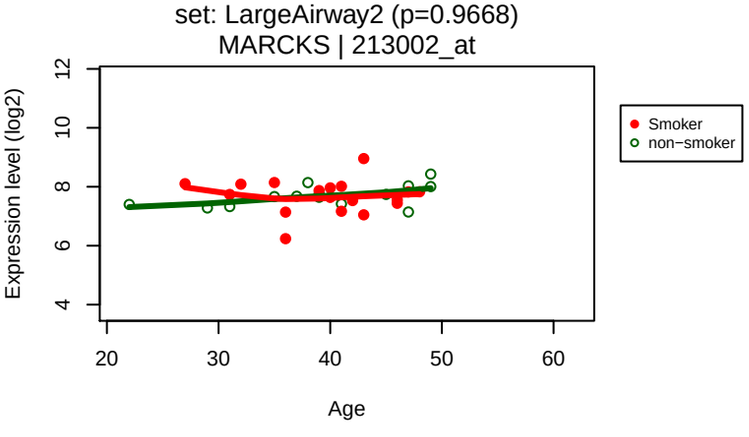

Gene expression ratio (smokers/nonsmokers)
gene: MARCKS | 213002_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3967
p=0.4393
p=0.8707
p=0.8212
p=0.9668
p=0.0268
p=0.0028
p=0.0253
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

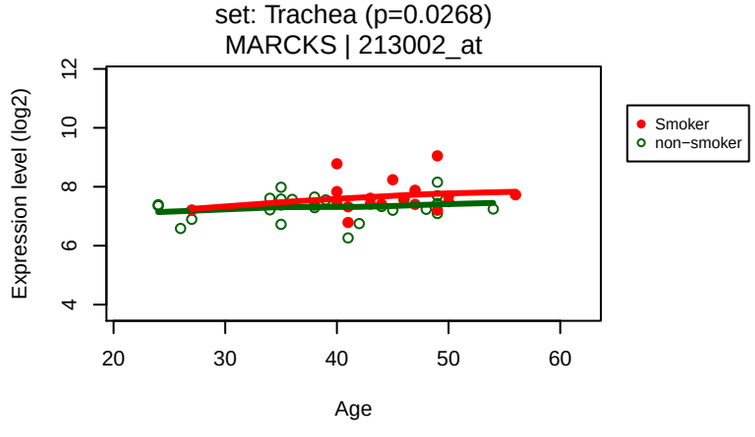

Gene expression ratio in Males
gene: MARCKS | 213002_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6398
p=0.595
p=0.0984
p=0.0249
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MARCKS | 213002_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5596
p=0.2592
p=0.0621
p=0.6783
SmallAirway2
LargeAirway2
Trachea
Macrophage2

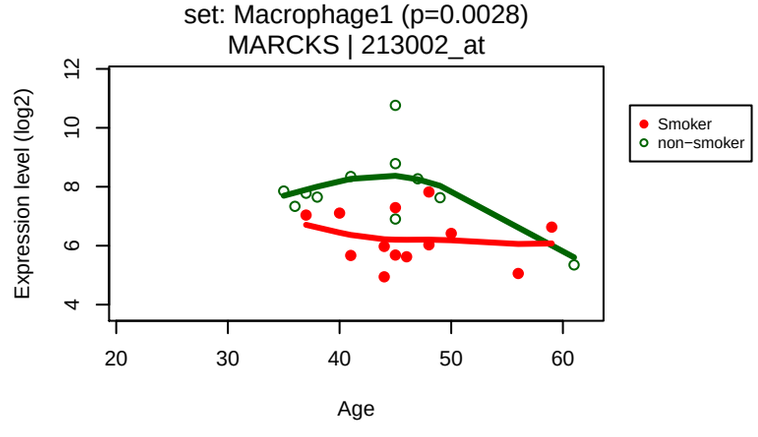

set: SmallAirway1 (p=0.3967)
MARCKS | 213002_at
Expression level (log2)
Age
Smoker
non−smoker

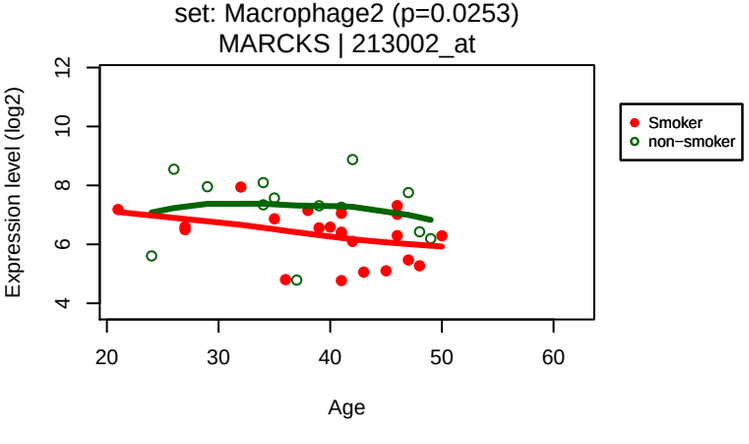

set: SmallAirway2 (p=0.4393)
MARCKS | 213002_at
Expression level (log2)
Age
Smoker
non−smoker

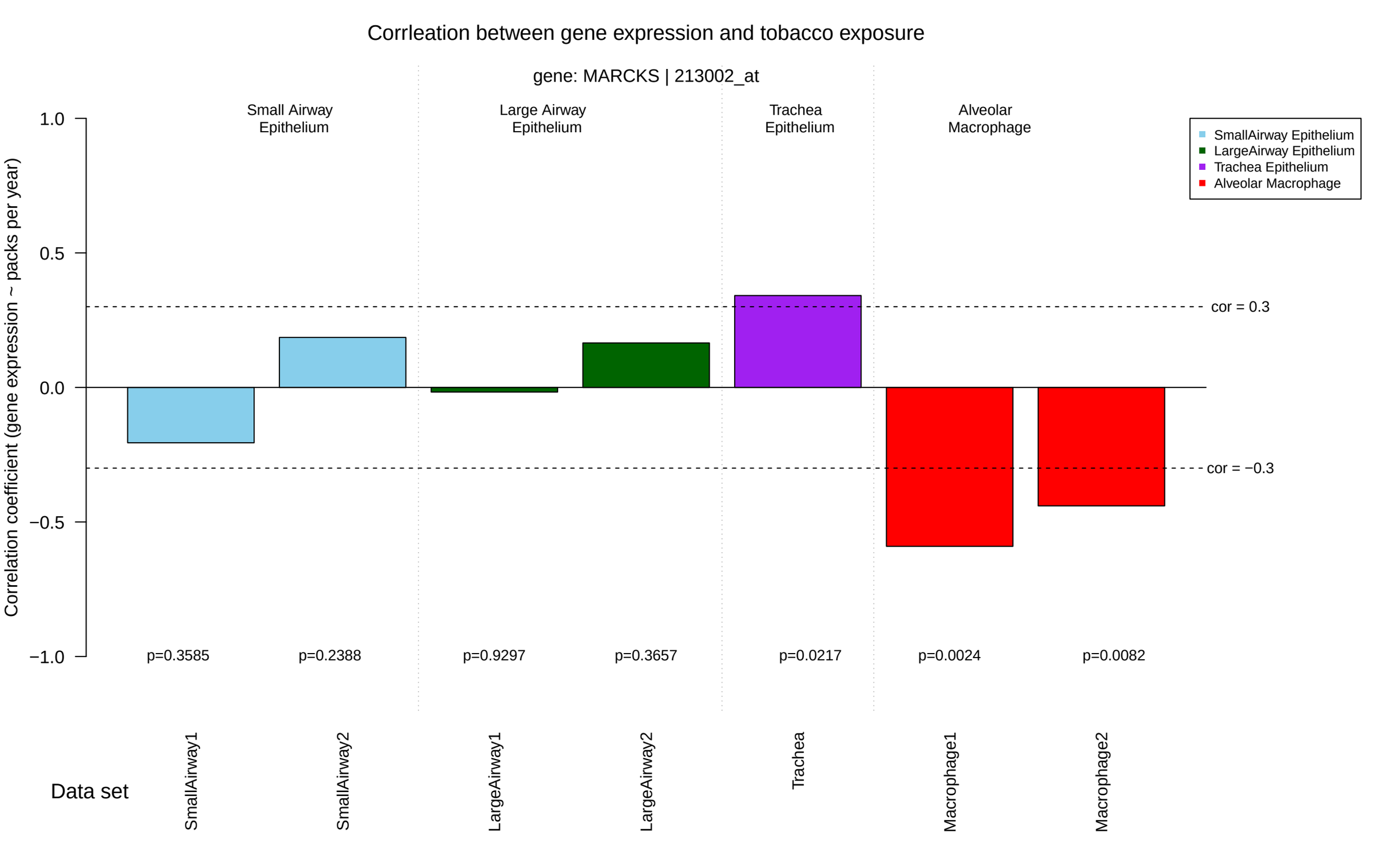

set: LargeAirway1 (p=0.8212)
MARCKS | 213002_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9668)
MARCKS | 213002_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0268)
MARCKS | 213002_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0028)
MARCKS | 213002_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0253)
MARCKS | 213002_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MARCKS | 213002_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3585
p=0.2388
p=0.9297
p=0.3657
p=0.0217
p=0.0024
p=0.0082
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2